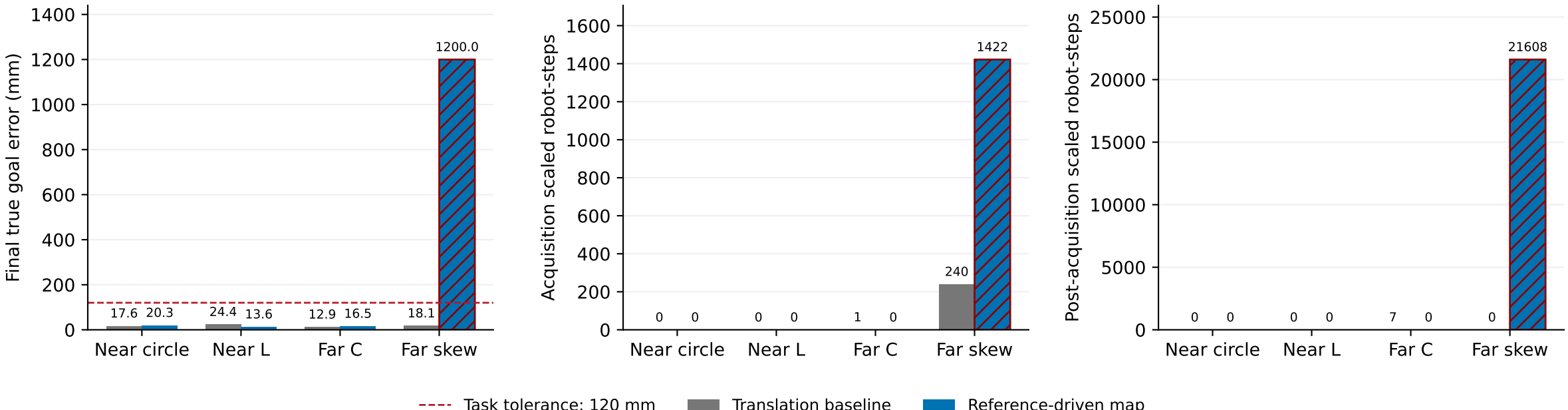
Reference-driven map: complete paired development outcomes
Final true goal error (mm)
1400
1200
1000
800
600
400
200
0
17.6
20.3
24.4
13.6
12.9
16.5
18.1
1200.0
Near circle
Near L
Far C
Far skew
Acquisition scaled robot-steps
1600
1400
1200
1000
800
600
400
200
0
0
0
0
0
1
0
240
1422
Near circle
Near L
Far C
Far skew
Post-acquisition scaled robot-steps
25000
20000
15000
10000
5000
0
0
0
0
0
7
0
0
21608
Near circle
Near L
Far C
Far skew
Task tolerance: 120 mm
Translation baseline
Reference-driven map
Eight fresh episodes on four reused conditions. Hatched bar: failed safe completion. All numerical parameters preserved.